Practice workbook · Trace tables: run the algorithm by hand

Executing a supplied flowchart, pseudocode algorithm or short program by hand for given inputs to find the outputs, the final or returned value of a variable, or to fill a trace table or output table.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T1-01

1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

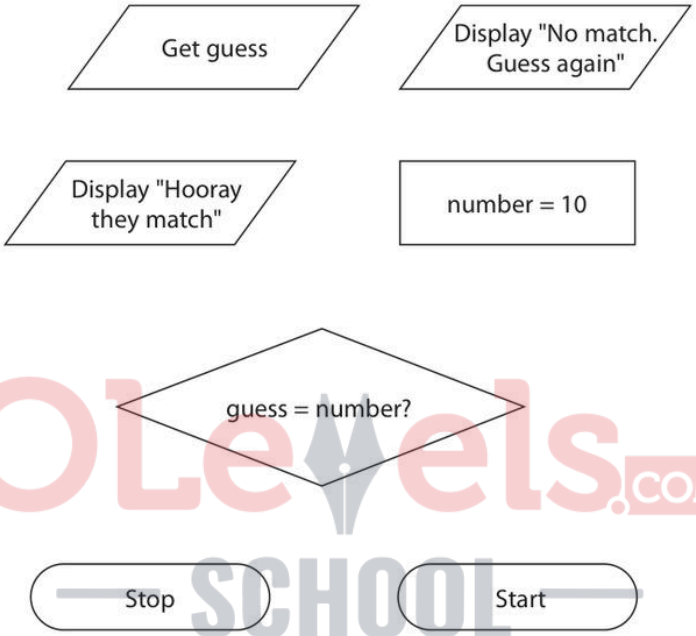
Worked-example retry · Attempt part c.i only.

4 Reba likes writing programs.

(a) She is writing a guessing game.

She needs a flowchart to show the logic of the game.

(i) These are the components needed to draw the flowchart.



Draw the flowchart for the algorithm in the box on the next page.

Use each component once.

Do not add any additional components.

Use as many arrows and yes/no labels as you need.

(5)

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10

P 7 2 5 3 8 A 0 1 0 2 0

1. Question 4 · part c.i · 4 marks

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Worked-example retry · Attempt part c.i only.

DO NOT WRITE IN THIS AREA

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Draw your flowchart here.

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1. Question 4 · part c.i · 4 marks

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Worked-example retry · Attempt part c.i only.

(ii) Identify an alternative method for writing the algorithm.

(1)

- ☒ **A** Simulation
- ☒ **B** Cipher
- ☒ **C** Program code
- ☒ **D** Truth table

(b) Reba wants to develop a program that will convert a temperature in Fahrenheit to Celsius.

Here are four steps in the algorithm.

The steps are not in the correct order.

	Step
A	Change the temperature to Celsius
B	Get the temperature in Fahrenheit
C	Show the temperature in Celsius
D	Set the temperature to 0

(i) Give the letter of the step that initialises a variable.

(1)

(ii) Give the letter of the step that inputs a value.

(1)

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P 7 2 5 3 8 A 0 1 2 2 0

1. Question 4 · part c.i · 4 marks

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Worked-example retry · Attempt part c.i only.

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QUESTION 4 (c) BEGINS ON THE NEXT PAGE.

1. Question 4 · part c.i · 4 marks

2022 JUN · Paper 01

Worked-example retry · Attempt part c.i only.

(c) **Figure 2** shows the pseudocode for an early version of an algorithm that Reba has written for another game.

The algorithm:

- asks the user to input a colour or input –1 to end the game
- awards 1 point for red
- awards 8 points for orange
- generates the score for the game
- displays the results of the game.

```
SET Colour TO ""
SET Score TO 0
SET RedPoints TO 0
SET OrangePoints TO 0
SET NumOranges TO 0

WHILE Colour <> "-1" DO
   RECEIVE Colour FROM (STRING) KEYBOARD
   IF Colour = "red" THEN
      SET RedPoints TO RedPoints + 1
   ELSE
      IF Colour = "orange" THEN
         SET OrangePoints TO OrangePoints + 8
         SET NumOranges TO NumOranges + 1
      END IF
   END IF
END WHILE

SET Score TO RedPoints + OrangePoints

SEND ("Score: "& Score) TO DISPLAY
SEND ("Number of reds: "& RedPoints) TO DISPLAY
SEND ("Number of oranges: "& OrangePoints) TO DISPLAY
```

Figure 2

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DO NOT WRITE IN THIS AREA

P 7 2 5 3 8 A 0 1 4 2 0

1. Question 4 · part c.i · 4 marks

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Worked-example retry · Attempt part c.i only.

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Reba inputs: red, orange, red, red, orange, –1

The outputs are not as she expects.

(i) Complete the trace table to show the outputs.

(4)

Colour	Score	RedPoints	OrangePoints	NumOranges	Outputs
	0	0	0	0	
red					
orange					
red					
red					
orange					
–1					

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(ii) Give the line number of the pseudocode that contains the error.

(1)

.....

(iii) Write a replacement line of pseudocode to correct the error.

(1)

.....

(Total for Question 4 = 14 marks)

1. Question 4 · part c.i · 4 marks · Working and answer

Working and answer

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2. Question 5 · part a.i · 3 marks

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

5 Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

Figure 2 shows an algorithm written using flowchart symbols.

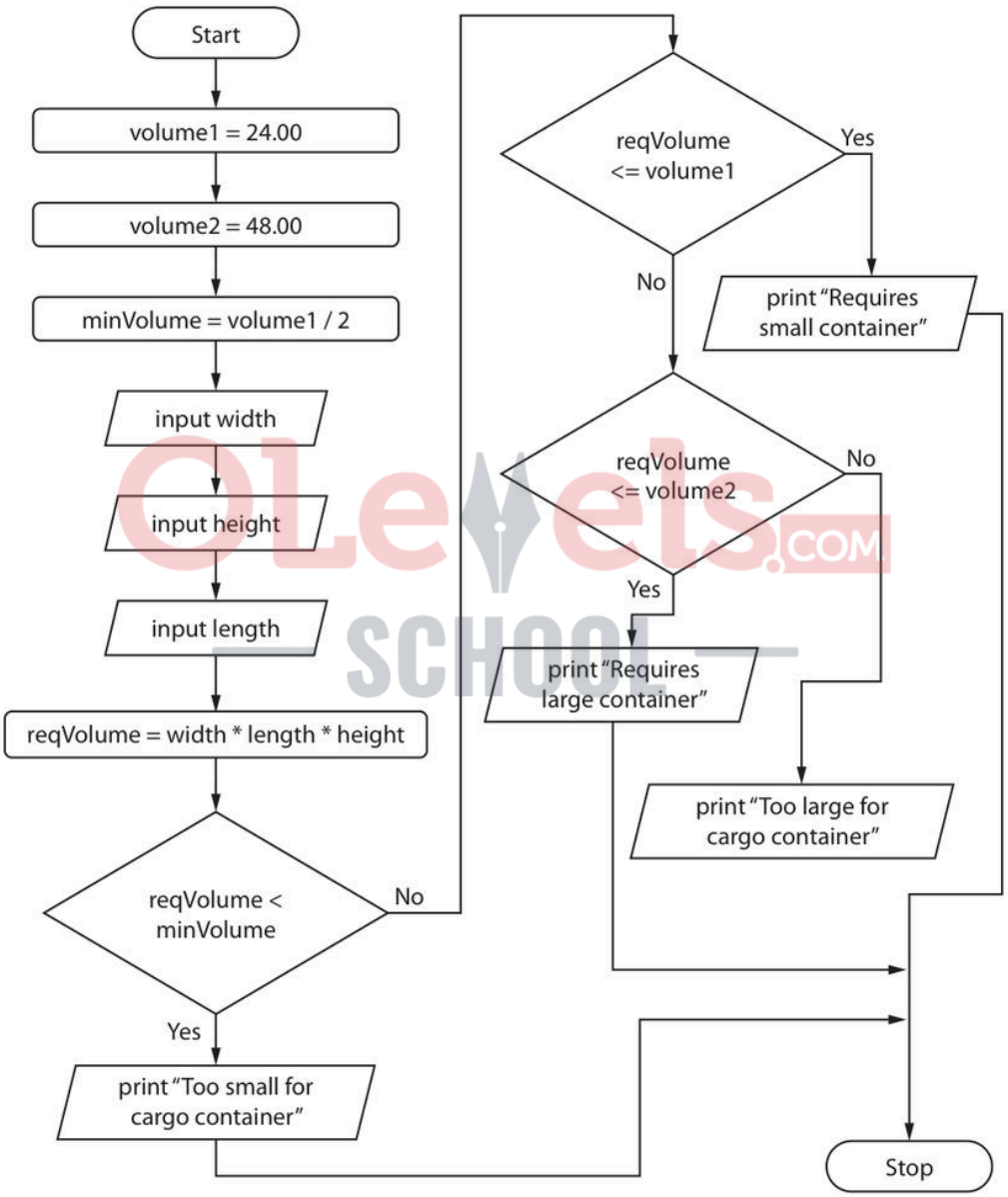


Figure 2

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2. Question 5 · part a.i · 3 marks

2019 JUN · Paper 01

Further practice · Attempt part a.i only.

(i) Complete the table to show the output for each cargo item.

(3)

Cargo item			Output
width	length	height	
4	4	2	
2	2	2	
3	8	5	

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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2. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

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3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

5 Ships carry cargo around the world in containers.

(a) Containers come in two sizes.

Figure 2 shows an algorithm written using flowchart symbols.

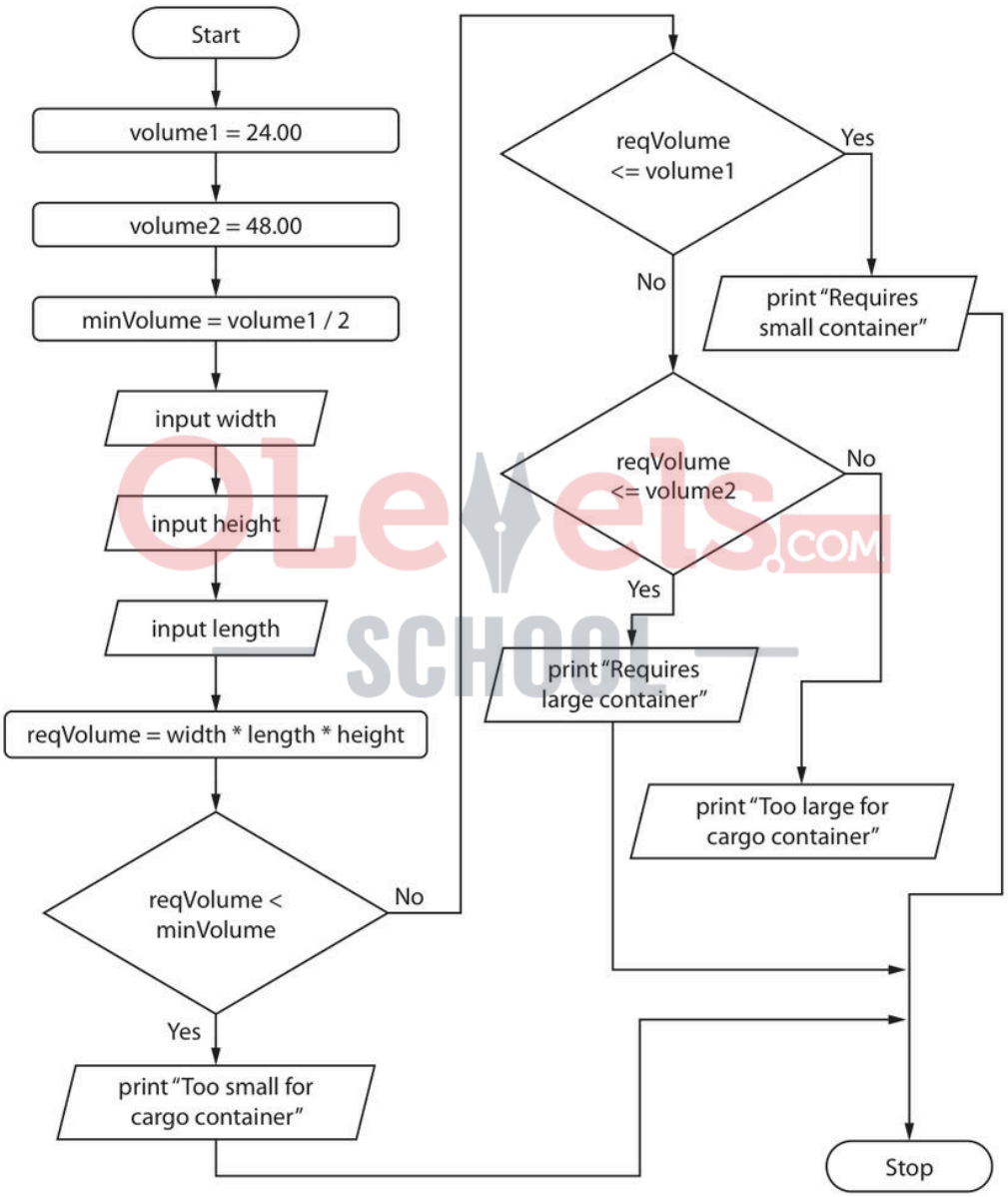


Figure 2

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3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

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(i) Complete the table to show the output for each cargo item.

(3)

Cargo item			Output
width	length	height	
4	4	2	
2	2	2	
3	8	5	

(ii) State the purpose of the algorithm in **Figure 2**.

(1)

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3. Question 5 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

(b) Each ship is registered to a state.

Figure 3 shows a partially completed algorithm written in pseudocode.

The completed algorithm must:

- print each state to the display on a new line
- count the number of states
- create a message as a single string (e.g. there are *number* states)
- print the message to the display.

Complete the algorithm in the space provided in **Figure 3**.

(4)

```
SET numStates TO 0
SET states TO ["France", "Singapore", "Malta", "Panama", "Greece", "Italy"]

FOR EACH state FROM states DO




END FOREACH
```

Figure 3

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA

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3. Question 5 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

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3. Question 5 · part c.i · 2 marks

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Further practice · Attempt part c.i only.

(c) Cargo ships have maximum weight loads.

Figure 4 shows an algorithm written using pseudocode.

The algorithm should identify the size of cargo ship required for any load.

There is an error on line 9.

```
2   SET loadWeight TO [20000, 28000, 40000, 50000]
3   SET index TO 0
4   SET found TO FALSE
5
6   SEND "Enter cargo weight" TO DISPLAY
7   RECEIVE target FROM (INTEGER) KEYBOARD
8
9   WHILE (NOT found) DO
10      IF (loadWeight [index] >= target) THEN
11          SEND loadWeight [index] TO DISPLAY
12          SET found TO TRUE
13      ELSE
14          SET index TO  index + 1
15      END IF
16  END WHILE
17
18  IF (NOT found) THEN
19      SEND "No ship available" TO DISPLAY
20  END IF
```

Figure 4

DO NOT WRITE IN THIS AREA
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DO NOT WRITE IN THIS AREA

P 6 1 8 8 0 A 0 1 8 2 4

3. Question 5 · part c.i · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c.i only.

(i) Trace tables are used to identify errors in algorithms.

Complete the trace table for an input of 50500 to show what happens due to the error on line 9 in the pseudocode in **Figure 4**.

You may not need to fill in all the rows in the table.

(2)

target	found	index	loadWeight[index]
50500	FALSE	0	

(ii) Construct a single line of pseudocode to correct line 9.

(2)

.....

.....

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.....

(Total for Question 5 = 12 marks)

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3. Question 5 · part c.i · 2 marks · Working and answer

Working and answer

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4. Question 2 · part b.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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4. Question 2 · part b.i · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.i only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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4. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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5. Question 2 · part b.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.iv only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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5. Question 2 · part b.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.iv only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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5. Question 2 · part b.iv · 1 mark

2019 JUN · Paper 02

Further practice · Attempt part b.iv only.

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(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

(2)

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5. Question 2 · part b.iv · 1 mark · Working and answer

Working and answer

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.

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2 Anna has a 5-year-old daughter, Beatrice, and a 14-year-old son, Graham.

(a) Anna wants to program a guessing game for Beatrice to play on the computer.

The program will generate a random number between 1 and 10.

Beatrice has to guess the number.

This pseudocode contains the logic required to create the game.

```
1   #   Initialise variables
2       SET counter TO 1
3       SET answer TO RANDOM(9) + 1     # i.e. a random integer 1 to 10 inclusive
4       SET guess TO 0
5
6   #   Print prompt and take guess from user
7       SEND "Enter a number from 1 to 10: " TO DISPLAY
8       RECEIVE guess FROM (INTEGER) KEYBOARD
9
10  #   Create while loop to check guess
11      WHILE guess <> answer DO
12          SET counter TO counter + 1
13          IF guess > answer THEN
14              SEND (guess & " was too high. Try again.") TO DISPLAY
15          ELSE
16              SEND (guess & " was too low. Try again.") TO DISPLAY
17          END IF
18          SEND "Guess again: " TO DISPLAY
19          RECEIVE guess FROM (INTEGER) KEYBOARD
20      END WHILE
21
22  #   Report the correct answer to the user and display the number of guesses
23      SEND ("You guessed " & guess & " in " & counter & "guesses.") TO DISPLAY
```

Write a program to implement the logic in the pseudocode.

Open **Q02a** in the code editor.

You **must** use the structure give in **Q02a** to write the program.

Do not add any further functionality.

Save your code as **Q02aFINISHED** with the correct file extension for the programming language.

(10)

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.

(b) Anna has written a program to help Graham with calculations for his science lessons.

Open **Q02b** in the code editor.

Use the code to answer these questions.

(i) State the value of **distance** when the program finishes.

(1)

(ii) Give **one** reason why there are brackets around **newSpeed + initialSpeed**

(1)

(iii) State the data type for the variable **distance**.

Justify your answer.

(2)

Data type

Justification

(iv) State the value of **distance** when **acceleration** is 2.75

(1)

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(c) Anna has written a program that manipulates text.

Open **Q02c** in the code editor.

Use the code to answer these questions.

(i) Identify a line number where the code includes a relational operator.

(1)

(ii) Identify a line number where a variable is set to a numeric value entered by a user.

(1)

(iii) Identify the name of a variable passed from the main program to the subprogram.

(1)

(iv) Identify the name of a variable that receives the value calculated by the subprogram.

(1)

(v) State the value returned by the subprogram when the input values are **educational**, **3**, **6**

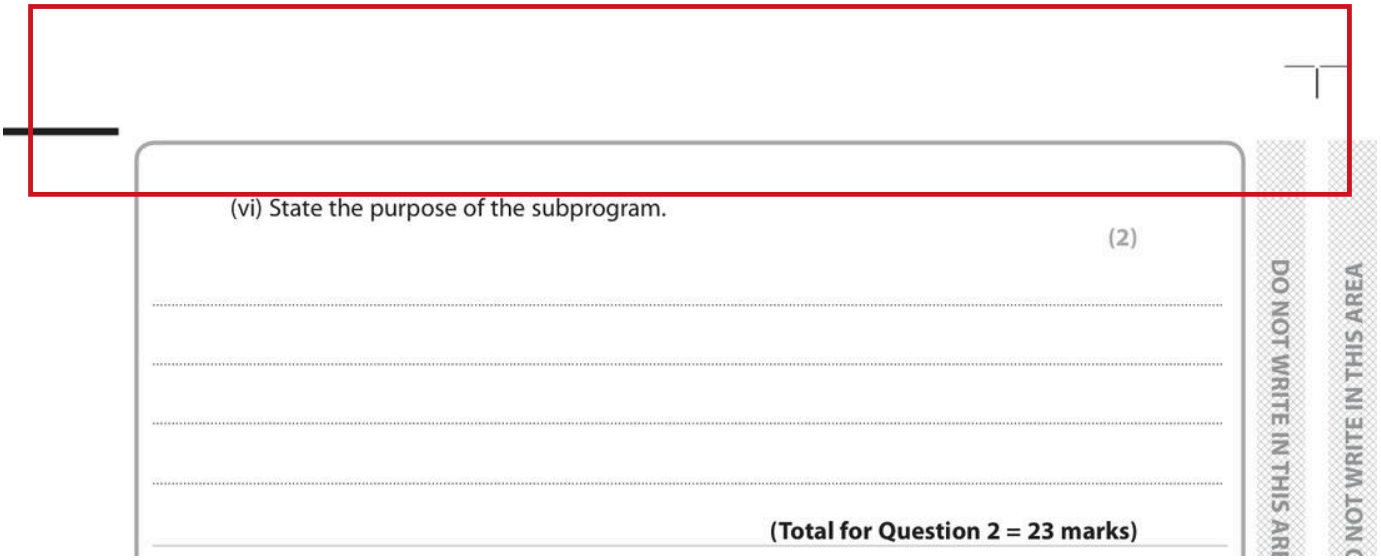
(2)

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6. Question 2 · part c.v · 2 marks

2019 JUN · Paper 02

Further practice · Attempt part c.v only.


(vi) State the purpose of the subprogram.

(2)

(Total for Question 2 = 23 marks)

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Working and answer

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7. Question 6 · part b.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

6 A software engineer is working with some algorithms.

(a) An algorithm needs to be developed.

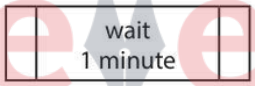
The algorithm must:

- check for a change of hour every minute
- use the library subprogram getHour() to get the hour part of the current time in the 24-hour clock (0 to 23)
- output "Good morning" when the hour is between 3 and 12, inclusive
- output "Good afternoon" when the hour is between 13 and 19, inclusive
- output "Good night" at all other times.

Complete the flowchart to represent this algorithm, in the space provided on the next page.

The call to the library subprogram getHour() has already been included.

Use this subprogram symbol to show waiting for 1 minute.



(6)

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7. Question 6 · part b.i · 2 marks

2020 NOV · Paper 01

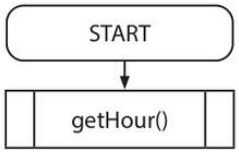
Further practice · Attempt part b.i only.

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Draw your flowchart here.



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7. Question 6 · part b.i · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

(b) Another algorithm determines when to change the flavoured syrups for a drinks dispensing machine.

Figure 4 shows this algorithm written in pseudocode.

```
1  SET flavours TO ["anise", "mango", "cola", "apple", "papaya", "strawberry",
   "lychee", "banana"]
2
3  SET volume TO [0.7, 0.2, 0.6, 0.1, 0.05, 0.8, 0.4, 0.6]
4
5  SET i TO 0
6
7  WHILE i < 8 DO
8
9      IF (volume[i] < 0.1) THEN
10         SEND (flavours[i] & " needs changing") TO DISPLAY
11
12     ELSE
13         IF (volume[i] >= 0.3) AND (volume[i] <= 0.5) THEN
14             SEND (flavours[i] & " needs ordering")  TO DISPLAY
15         END IF
16     END IF
17
18     SET i TO i + 1
19
20 END WHILE
```

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Figure 4

(i) Give the output produced by the algorithm.

(2)

.....

.....

.....

.....

The algorithm works with a fixed number of flavours.

(ii) Give the number of the line in the algorithm that would need to be amended to allow for any number of flavours.

(1)

.....

.....

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7. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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8. Question 5 · part a.i · 5 marks

2021 JUN · Paper 01

Further practice · Attempt part a.i only.

5 Isaac is a program developer.

(a) **Figure 3** shows an algorithm Isaac has written.

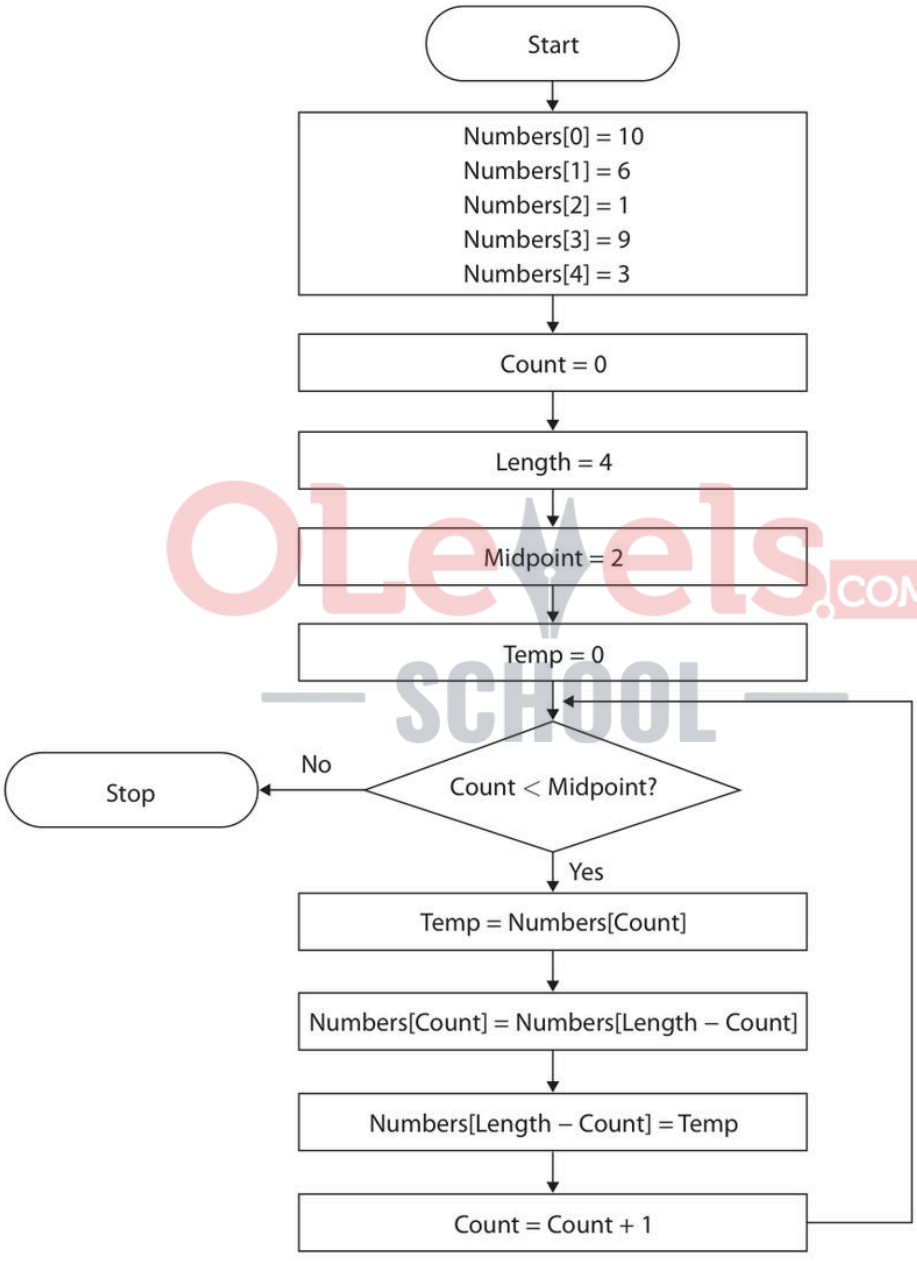


Figure 3

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8. Question 5 · part a.i · 5 marks

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Further practice · Attempt part a.i only.

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(i) Complete the trace table. You may not need to use all of the rows.

(5)

				Numbers array				
Count	**Length**	**Midpoint**	**Temp**	[0]	[1]	[2]	[3]	[4]
0	4	2	0	10	6	1	9	3

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(ii) Give the purpose of the algorithm.

(1)

(iii) Explain why the variable Temp is needed.

(2)

8. Question 5 · part a.i · 5 marks · Working and answer

Working and answer

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9. Question 3 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

3 Several encryption algorithms have been developed.

(a) Ahmed has started to create a Caesar cipher encryption program.

Caesar cipher encryption works by giving a number value to a key.

Each letter in a plaintext message is replaced by a new ciphertext letter using the key.

For example, as shown in the table, a key value of +2 would change the plaintext message **encrypt** to the ciphertext message **gpetarv**.

Plaintext	e	n	c	r	y	p	t
Ciphertext	g	p	e	t	a	r	v

Ahmed's program already converts from plaintext to ciphertext.

He wants to extend the program to:

- accept the input of a plaintext message
- accept the input of the key, which must be a positive integer between 1 and 25
- write the generated ciphertext to a text file named **Cipher.txt**
- display the ciphertext.

Open **Q03a** in the code editor.

Complete the program code.

You must use the structure given in the file **Q03a** to complete the program.

Do not add any further functionality.

Save your code as **Q03aFINISHED** with the correct file extension for the programming language.

(8)

(b) Explain **one** reason why the Caesar cipher is a weak encryption algorithm.

(2)

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(c) The Vigenere cipher encrypts plaintext using a series of Caesar ciphers, based on the letters of a keyword and a Vigenere square.

Figure 1 shows a Vigenere square.

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

Complete the table to perform a Vigenere decryption of the ciphertext message **flmktrwhee** using the keyword **thirty**.

(4)

Encrypted letter	f	l	m	k	t	r	w	h	e	e
Keyword letter										
Decrypted letter										

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9. Question 3 · part d.i · 1 mark

2021 JUN · Paper 02

Further practice · Attempt part d.i only.

(d) A user-defined function uses a Vigenere cipher to encrypt plaintext into ciphertext.

This pseudocode shows how the function is called.

```
SET ciphertext TO ciphertext & getCipherLetter(keywordLetter, plaintextLetter)
```

(i) State what would be returned by this function.

(1)

(ii) Give the name of **one** parameter passed to this function.

(1)

(iii) State what is meant by a **built-in subprogram**.

(1)

(Total for Question 3 = 17 marks)

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Working and answer

10. Question 4 · part b.i · 5 marks

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Further practice · Attempt part b.i only.

4 Algorithms can be used to perform calculations and to process data.

(a) State what is meant by the term **algorithm**.

(1)

.....

.....

(b) **Figure 3** shows an algorithm.

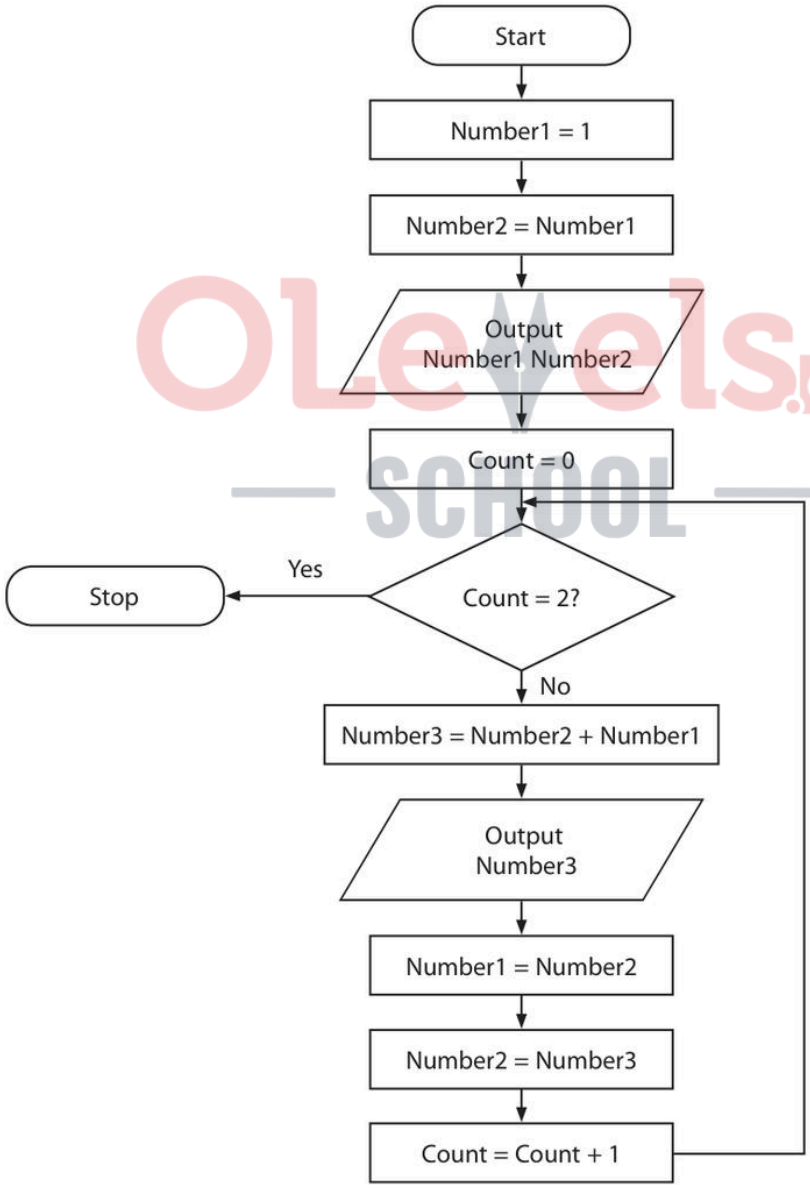


Figure 3

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P 6 9 2 9 7 A 0 1 2 2 0

10. Question 4 · part b.i · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

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(i) Complete this trace table for the algorithm.

You may not need to use all the rows.

(5)

Number1	Number2	Number3	Count	Output	Count = 2?
1	1	–	0	1 1	False

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(ii) The benefits of using a trace table include that they allow variable states, outputs and decisions to be recorded.

Give **one other** benefit of using a trace table to test an algorithm.

(1)

.....

.....

10. Question 4 · part b.i · 5 marks · Working and answer

Working and answer

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11. Question 3 · part b · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

3 Algorithms can be represented in flowcharts, pseudocode or program code.

(a) Trace tables can be used with flowcharts or pseudocode.

Give **two** characteristics of a trace table.

(2)

1 .....

.....

2 .....

.....

(b) An algorithm has been written to validate numbers entered by the user.

Figure 2 shows the pseudocode for the algorithm.

```
1  RECEIVE num1 FROM (INTEGER) KEYBOARD
2  RECEIVE num2 FROM (INTEGER) KEYBOARD
3  IF (((num1 < 16) AND (num2 < 23)) OR (num2 = 13)) THEN
4      SEND "State 1" TO DISPLAY
5  ELSE
6      IF ((num2 > 12) AND (num1 > 20)) THEN
7          SEND "State 2" TO DISPLAY
8      ELSE
9          IF ((num1 = 88) OR NOT (num2 = 18)) THEN
10             SEND "State 3" TO DISPLAY
11         ELSE
12             SEND "State 4" TO DISPLAY
13         END IF
14     END IF
15 END IF
```

Figure 2

Complete the table to show the output for each set of inputs.

(3)

num1	num2	Output
88	18	
17	18	
12	19	

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P 7 2 9 3 8 A 0 6 1 6

11. Question 3 · part b · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

(c) A program is required to calculate the result of raising one integer (the base) to the power of another (the exponent).

Figure 3 shows a flowchart for the algorithm.

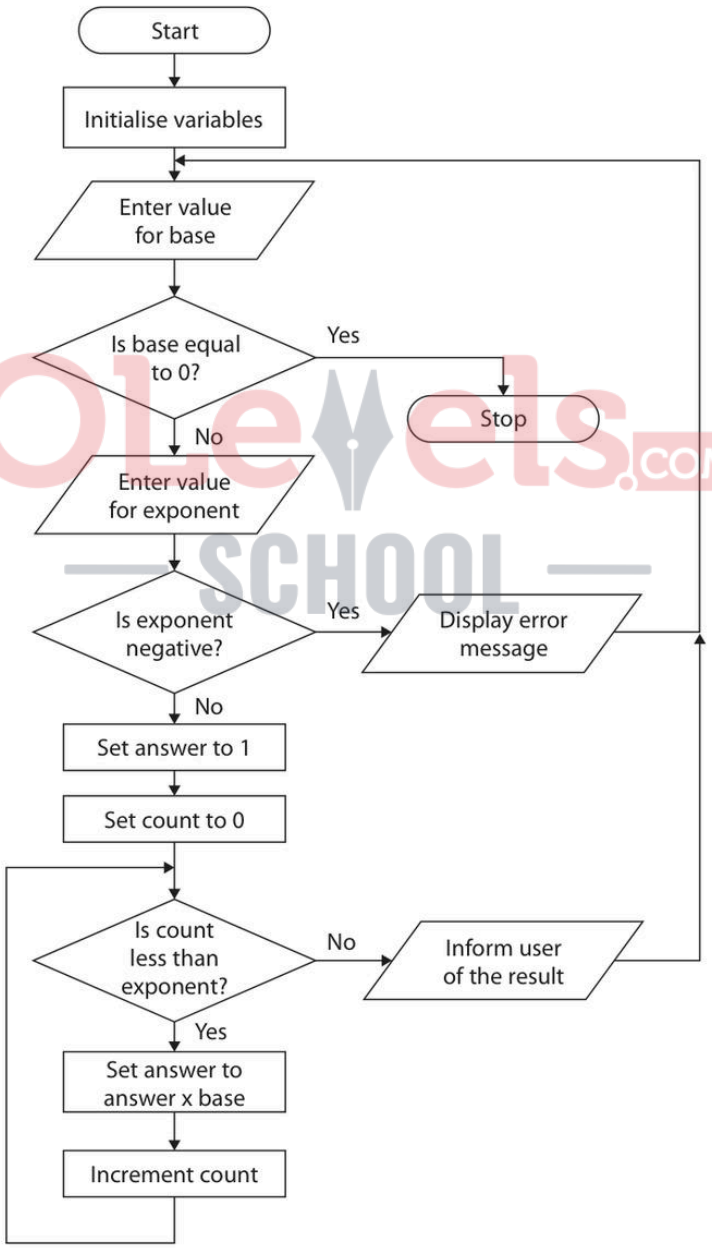


Figure 3

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11. Question 3 · part b · 3 marks · Working and answer

Working and answer

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12. Question 6 · part b.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

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12. Question 6 · part b.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

12. Question 6 · part b.i · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

(iii) State the purpose of the algorithm.

(1)

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

(Total for Question 6 = 13 marks)

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12. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

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13. Question 6 · part b.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

6 Algorithms can be represented using flowcharts or pseudocode.

(a) An algorithm has been designed to perform these steps:

- A user is asked to input a number **n**
- If the number **n** is 0 the program stops
- If the number **n** is positive a message to display 'Positive' is output
- If the number **n** is negative a message to display 'Negative' is output
- The program repeats until a value of 0 has been input.

Draw a flowchart to represent this algorithm.

(6)

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13. Question 6 · part b.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

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(b) **Figure 3** shows an algorithm written in pseudocode that performs a mathematical process.

```
01
02  RECEIVE number FROM (INTEGER) KEYBOARD
03
04  SET flag TO FALSE
05
06  IF number = 1 THEN
07     SET flag TO TRUE
08  ELSE
09     SET testNum TO 1
10     REPEAT
11        SET testNum TO testNum + 1
12        IF (number MOD testNum) = 0 THEN
13           SET flag TO TRUE
14        END IF
15     UNTIL (testNum = number) OR (flag = TRUE)
16  END IF
17
18  IF NOT (flag = TRUE)
19     SEND number TO DISPLAY
20  END IF
21
```

Figure 3

Trace tables are used to show the outputs from algorithms.

(i) Complete the trace table for an input of 7

(2)

number	flag	testNum	DISPLAY

13. Question 6 · part b.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.ii only.

(ii) Complete the trace table for an input of 9

(2)

number	flag	testNum	DISPLAY

(iii) State the purpose of the algorithm.

(1)

(iv) Lines 10 to 15 of the pseudocode in **Figure 3** use a post-conditioned loop.

Explain **one** reason why it is better for the algorithm to use a post-conditioned loop rather than a count controlled loop.

(2)

(Total for Question 6 = 13 marks)

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13. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

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